

# UX Research

An overview of purpose, methods, process, and value in product design

## 1. Introduction

User Experience (UX) research is the systematic study of users — their behaviors, needs, motivations, and pain points — through observation and feedback. Its goal is to replace assumptions with evidence so that products are useful, usable, and desirable. Instead of asking “What do we think users want?”, UX research asks “What do users actually do, and why?”

Research is not a single phase at the start of a project. It is applied throughout the product life cycle: before design to discover opportunities, during design to validate ideas, and after launch to measure and improve real-world performance. This is consistent with the human-centred design approach described in ISO 9241-210, which stresses early and continuous involvement of users.

## 2. Why UX Research Matters

- **Reduces risk:** problems found in research or prototype testing are far cheaper to fix than problems found after development and release.
- **Improves usability and satisfaction:** products designed around real tasks and mental models are easier to learn and use.
- **Supports better decisions:** teams can prioritize features using evidence instead of opinions or the loudest voice in the room.
- **Builds empathy:** direct contact with users helps designers, developers, and stakeholders understand the people they are building for.
- **Drives business results:** higher conversion, lower support costs, and better retention are common outcomes of fixing usability problems.

## 3. Types of UX Research

UX research methods are usually classified along two dimensions, a framework popularized by Christian Rohrer (Nielsen Norman Group).

### 3.1 Qualitative vs. Quantitative

**Qualitative** research answers *why* and *how*. It gathers rich, descriptive data from small samples (e.g., interviews, observation). **Quantitative** research answers *how many* and *how much*. It collects numeric data from larger samples (e.g., surveys, analytics, A/B tests) and supports statistical conclusions.

### 3.2 Attitudinal vs. Behavioral

**Attitudinal** research captures what people say — their opinions, beliefs, and stated preferences. **Behavioral** research captures what people *do*. Because the two often differ, strong research programs combine both.

## 4. Common UX Research Methods

| Method                           | Type                        | What it tells you                                | Typical use                                 |
|----------------------------------|-----------------------------|--------------------------------------------------|---------------------------------------------|
| User interviews                  | Qualitative, attitudinal    | Motivations, goals, frustrations, context of use | Discovery; understanding needs              |
| Surveys / questionnaires         | Quantitative, attitudinal   | Opinions and satisfaction at scale               | Measuring satisfaction; validating findings |
| Contextual inquiry / field study | Qualitative, behavioral     | How people work in their real environment        | Discovery; complex workflows                |
| Usability testing                | Qual. or quant., behavioral | Where users struggle completing real tasks       | Design validation; before and after launch  |

| Method               | Type                       | What it tells you                              | Typical use                                  |
|----------------------|----------------------------|------------------------------------------------|----------------------------------------------|
| Card sorting         | Qualitative / quantitative | How users group and label content              | Information architecture; navigation         |
| Diary studies        | Qualitative, mixed         | Behavior and experiences over time             | Long-term or recurring use cases             |
| A/B testing          | Quantitative, behavioral   | Which design variant performs better           | Optimizing live products                     |
| Analytics & heatmaps | Quantitative, behavioral   | Where users click, drop off, and spend time    | Finding problem areas; monitoring            |
| Heuristic evaluation | Expert review              | Violations of established usability principles | Quick, low-cost review of an existing design |

## 5. The UX Research Process

Although every project differs, most research follows a similar sequence:

- **1. Define objectives and research questions.** Clarify what the team needs to learn and what decision the research will inform. Vague goals produce vague findings.
- **2. Choose the method.** Select the method(s) that fit the questions, product stage, timeline, and budget.
- **3. Recruit participants.** Find people who represent the real target audience. Poor recruitment is one of the most common causes of misleading results.
- **4. Prepare materials.** Write the interview guide, survey, or task scenarios; prepare prototypes; run a pilot session.
- **5. Conduct the study.** Ask neutral, open-ended questions; observe more than you talk; get informed consent and protect privacy.
- **6. Analyze the data.** Code and cluster notes (for example, using affinity diagramming) to find patterns, then rank issues by severity and frequency.
- **7. Report and act.** Share clear insights and actionable recommendations, using personas, journey maps, or short highlight videos. Research only creates value when it changes decisions.

## 6. Key Deliverables

Common outputs include **personas** (archetypes based on real data), **user journey maps** (the steps, emotions, and pain points across an experience), **empathy maps**, **usability reports** with severity ratings, and **information architecture** recommendations such as sitemaps derived from card sorting.

## 7. Metrics and Measurement

UX can be measured. Frequently used measures include **task success rate**, **time on task**, **error rate**, the **System Usability Scale (SUS)** — a ten-item questionnaire developed by John Brooke in 1996 that yields a score from 0 to 100 — the **Net Promoter Score (NPS)**, and **customer satisfaction (CSAT)**. Pairing these numbers with qualitative insights explains not only *what* changed but *why*.

## 8. Best Practices and Common Pitfalls

- **Test early and often.** Small, frequent studies beat a single large one. Nielsen's well-known guidance suggests that about five users per round can reveal most major usability problems in a qualitative test, so it is better to run several small rounds and iterate.
- **Avoid leading questions.** "Don't you think this is easy to use?" biases the answer. Ask "How did you find completing that task?" instead.
- **Watch behavior, not just words.** People are poor at predicting or recalling their own behavior.

- **Beware of bias.** Confirmation bias, sampling bias, and the researcher's own assumptions can all distort findings.
- **Be ethical.** Obtain consent, anonymize data, explain how it will be used, and respect participants' right to withdraw.
- **Make findings usable.** Keep reports short, visual, and tied to specific recommendations.

## 9. Conclusion

UX research connects design decisions to real human needs. By combining qualitative and quantitative methods, and attitudinal and behavioral data, teams can understand users deeply, validate ideas before they are costly to change, and measure improvement over time. Organizations that treat research as a continuous practice, rather than a one-time checkbox, consistently build products that people find more useful and more enjoyable.

## References

- Rohrer, C. (2014). *When to Use Which User-Experience Research Methods*. Nielsen Norman Group.
- Nielsen, J. (2000). *Why You Only Need to Test with 5 Users*. Nielsen Norman Group.
- Nielsen, J. (1994). *10 Usability Heuristics for User Interface Design*. Nielsen Norman Group.
- Brooke, J. (1996). SUS: A “quick and dirty” usability scale. In *Usability Evaluation in Industry*. Taylor & Francis.
- ISO 9241-210:2019. *Ergonomics of human-system interaction — Human-centred design for interactive systems*.
- Hall, E. (2013). *Just Enough Research*. A Book Apart.